

FilmTree: A parallel corpus of film subtitles

Alexandra Nogina Siavash Hajbahrmanian Kira Tulchynska Natalia Levshina

Jessica K. Ivani Alena Witzlack-Makarevich

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1 Corpus description

This corpus extends the ParTree corpus developed by [Ebert et al. \(2023\)](#), which, in turn, builds on the ParTy corpus by [Levshina \(2016\)](#). We selected subtitles for two films, ‘Bridge of Spies’ and ‘The Godfather’, and performed a fully manual alignment in order to improve cross-linguistic correspondences between translations. The resulting corpus consists of 53 languages from 13 language families: 92 texts in total for two films. Of these, 68 texts were taken directly from ParTree, while 24 were added from other sources. 42 languages with available models were further automatically annotated following the Universal Dependencies (UD) framework (v2.17; [Zeman et al., 2025](#)) using UDPipe2 ([Straka, 2018](#)).

The corpus files are organised into three folders, following the structure implemented in the ParTree corpus. The `raw` folder contains the original subtitle files in SRT format. The `text` folder contains files that have been converted to TXT format. The `conllu` folder contains files in CoNLL-U format that have been annotated with Universal Dependencies. Inside each folder, the files are organised into subfolders by film title.

For each film text file, the filename includes the film title, the ISO 639-3 language code,¹ and the language Glottocode, e.g. `Godfather_eng_stan1293`.

The corresponding filenames of the CoNLL-U files additionally include the UDPipe model identifier after the Glottocode, e.g. `Godfather_eng_stan1293_english-ewt`.

2 Workflow

1. 68 translations in TXT format were obtained from the [ParTree](#) corpus:²
 - One of those 68 translations was subsequently modified and improved by the speaker: ‘Bridge of Spies’ in Arabic;
2. 14 translations were downloaded in SRT format from [Open Subtitles](#):
 - The files were converted to TXT format;
 - Opening/closing credits, empty lines, and other non-dialogue elements were removed;
3. Ten translations were commissioned from the speakers:

¹Some of the ISO codes used in the ParTree naming conventions have been changed because they represented macrolanguages combining several language varieties. For example, FilmTree now uses `arb` for (Modern) Standard Arabic, in place of `ara`, which was previously used in ParTree for the Arabic macrolanguage.

²The total number of files available in ParTree for ‘The Godfather’ and ‘Bridge of Spies’ is higher; however, some translations have been judged as low quality, possibly due to machine translation. Where available, they were substituted with better versions of the text, while others were discarded from this corpus.

- ‘The Godfather’: in Armenian, Khoekhoe, Ruuli, Suansu, and Tangkhul Naga;
 - ‘Bridge of Spies’: in Khoekhoe, Luganda, Ruuli, Suansu, and Tangkhul Naga;
4. All subtitles were manually aligned using English as the pivot language³ (see Section 2.1 for a detailed description of the alignment process).
 5. A tagging system was implemented in a separate column to mark problematic lines.
 6. Subtitle files were minimally preprocessed for UD parsing based on alignment tags, such as removal of missing translations, handling of duplicated and split lines.
 7. The preprocessed subtitle files were then automatically annotated with Universal Dependencies (UD v2.17) using UDPipe2. Each language was parsed with the latest available model, and the resulting CoNLL-U outputs were merged and aligned back to the corpus structure.

2.1 Alignment process

1. The subtitles were aligned at the clause level, not at the sentence level. Specifically, the alignment aimed to ensure that each line corresponding to the English version, which has one sentence per line, contains all the equivalent clauses in the target language. As a result, some lines in the target languages contain more than one complete sentence, while others consist of units smaller than a full sentence (see the example in Table 1).

Table 1: Clausal correspondence between languages in ‘The Godfather’

ID	English	Portuguese
5	She found a boyfriend, not an Italian.	Ela arranhou um namorado. Não é italiano.
6	She went to the films with him.	Foi ao cinema com ele,
7	She stayed out late.	regressou tarde.
8	I didn’t protest.	Não protestei.

2. If a single expression in a translation corresponded to two or more lines in the English reference text and could not be meaningfully segmented, it was repeated across all corresponding lines (see the example in Table 2).

Table 2: Two-to-one correspondence between languages in ‘Bridge of Spies’

ID	English	Malay
79	Well, look, Bob.	Dengar sini, Bob.
80	May I?	Boleh saya panggil awak Bob?
81	Bob?	Boleh saya panggil awak Bob?

Such instances may pose a challenge for frequency-based analyses. To facilitate their identification, these lines were marked with a *forced duplication* tag in a separate column (see the example in Table 3).

³The English subtitles available in ParTree were slightly modified. This resulted in a different total number of lines. Some lines were merged in order to maintain clausal consistency: for example, in those cases where two lines clearly contained a single sentence that was too long to be represented in a single subtitle frame. Conversely, we split other lines into two since they contained two sentences that had been mistakenly joined together due to punctuation inconsistencies. Finally, the English text for ‘The Godfather’ was extended by three additional lines found in a different subtitle version on Open Subtitles.

Table 3: Use of the *forced duplication* tag in ‘Bridge of Spies’

ID	Malay	Malay_Tag
79	Dengar sini, Bob.	
80	Boleh saya panggil awak Bob?	forced duplication
81	Boleh saya panggil awak Bob?	forced duplication

3. If two or more sentences or clauses in the translation appeared in an order different from that of the English reference, their order was manually adjusted to ensure consistency with the English source (see the example in Table 4). In several cases, this reordering required minor edits to the text.

Table 4: Opposite clause order across languages in ‘The Godfather’

ID	English	Russian
<i>Before</i>		
1036	We’ve heard from Santino in New York...	Твои враги знают, что ты здесь.
1037	Your enemies know you’re here.	Мы узнали это от Сантино в Нью-Йорке.
<i>After</i>		
1036	We’ve heard from Santino in New York...	Мы узнали от Сантино в Нью-Йорке...
1037	Your enemies know you’re here.	Твои враги знают, что ты здесь.

4. In some cases, the translations contained extra text with no counterpart in the English subtitles. These typically included sentences spoken in other languages in the films: Italian in ‘The Godfather’, German and Russian in ‘Bridge of Spies’. Those lines were often not represented in the English subtitles, but were translated into other languages. Such instances were retained, grouped on a single line and enclosed in angle brackets <.../>. Additionally, a *missing English* tag was added in a separate annotation column (see the example in Table 5).

Table 5: Text without an English counterpart and use of the *missing English* tag in ‘The Godfather’

ID	English	French	French_Tag
448	I’m Bruno Tattaglia.	Je suis Bruno Tattaglia.	
449	I know.	Je sais. <Un scotch? D’avant-guerre! Je bois pas./>	missing English
450	You know who I am?	Tu me connais?	

5. All empty lines in the translations were cross-checked against the original SRT files from ParTree (i.e. the raw data prior to automatic alignment and enumeration, applied in both ParTy and ParTree). This step was necessary because automatic processing occasionally led to the loss of lines or partial content. Missing text was restored where possible. After validation, genuinely empty lines were marked with the [NO TRANSLATION] label, and a *missing translation* tag was placed in the separate tag column (see the example in Table 6).

Table 6: Empty lines in translations and use of the *missing translation* tag in ‘Bridge of Spies’

ID	English	Basque	Basque_Tag
589	So you’re not curious?	Ez al duzu jakinminik?	
590	No, not really.	[NO TRANSLATION]	missing translation
591	I always assumed you were an artist.	Artistatzat hartu zaitut beti.	

2.2 UD Parsing

1. The UD annotation was carried out for 42 languages with available models via the [UDPipe web service](#), using a Python-based UDPipe2 wrapper.
2. Prior to parsing, the data were minimally preprocessed based on alignment tags. Lines marked as *missing translation* were removed. Instances of *forced duplication* were collapsed into a single line with merged alignment IDs. Lines marked as *missing English* were split into subparts and reindexed to preserve alignment information. Minor orthographic normalization was applied where required by specific UD models.
3. Each file was parsed separately using the latest available UDPipe model for the respective language. We used version 2.17 models ([Zeman et al., 2025](#)), see model evaluation in the [UDPipe 2 documentation](#). The parsing was performed with each aligned subtitle line treated as a separate parsing unit and included tokenization, part-of-speech tagging, morphological analysis, and dependency parsing. The output was produced in CoNLL-U format.
4. Sentence identifiers (`sent_id`) were reassigned to match the corpus alignment scheme and alignment information was encoded in the MISC column using the Ref attribute.

3 Data

Table 7 contains film information (year, duration, genre) according to [IMDb](#) and the number of sentences and lines in the English subtitles in FilmTree.

Table 7: Film data

Film	Year	Duration (min)	Genre	Sentences/lines
Bridge of Spies	2015	142	Drama, History, Thriller	1885
The Godfather	1972	175	Crime, Drama	1769

Table 8 presents an overview of all the languages and texts available in FilmTree, grouped by their respective language families. Cells in [pale green](#) indicate texts taken from ParTree. Cells in [pale blue](#) indicate texts added to FilmTree from other sources: either translated by the speakers (or modified, as is the case with ‘Bridge of Spies’ in Arabic) or downloaded from OpenSubtitles. White cells indicate translations that are not present in the corpus either because they were unavailable or because the quality of the available translation was considered insufficient (e.g. machine-translated texts). The UD column shows whether a UD-annotated file is available for the language, see Section 2.2.

Table 8: Overview of the languages in FilmTree

Family	Language	Godfather	Bridge of Spies	UD
Afro-Asiatic	Arabic			parsed
	Hebrew			parsed
Atlantic-Congo	Luganda			not parsed
	Ruuli			not parsed
Austronesian	Indonesian			parsed
	Malay			not parsed
	Vietnamese			parsed
Dravidian	Malayalam			not parsed
	Telugu			parsed
Indo-European	Albanian			parsed
	Armenian			parsed
	Assamese			not parsed
	Bengali			not parsed
	Bosnian			not parsed
	Bulgarian			parsed
	Croatian			parsed
	Czech			parsed
	Danish			parsed
	Dutch			parsed
	English			parsed
	French			parsed
	German			parsed
	Greek			parsed
	Hindi			parsed
	Icelandic			parsed
	Italian			parsed
	Kurmanji			not parsed
	Lithuanian			parsed
	Macedonian			not parsed
	Norwegian			parsed
	Persian			parsed
	Polish			parsed
	Portuguese			parsed
	Romanian			parsed
	Russian			parsed
	Serbian			parsed
	Sindhi			parsed
	Slovak			parsed
	Slovenian			parsed
	Spanish			parsed
	Swedish			parsed
	Ukrainian			parsed
Isolate	Basque			parsed

Family	Language	Godfather	Bridge of Spies	UD
Japonic	Japanese			parsed
Khoe-Kwadi	Khoekhoe			parsed
Koreanic	Korean			parsed
Sino-Tibetan	Suansu			not parsed
	Tangkhul Naga			not parsed
Tai-Kadai	Thai			parsed
Turkic	Turkish ⁴			parsed
Uralic	Estonian			parsed
	Finnish			parsed
	Hungarian			parsed

Contributions

- Alexandra Nogina: collecting data for FilmTree, supervising contributors, aligning the texts, managing the project documentation;
- Siavash Hajbahramian: collecting data for FilmTree, supervising contributors, aligning the texts, contributing to coding and UD parsing, editing the project documentation;
- Kira Tulchynska: coding, UD parsing, editing of the project documentation;
- Natalia Levshina: collecting the original ParTy corpus, advising on data collection;
- Jessica K. Ivani: editing of the project documentation and consulting;
- Alena Witzlack-Makarevich: supervising contributors, consulting and funding.

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- Nane Avetisyan: translating ‘The Godfather’ into Armenian;
- Richard Selwanga: translating ‘Bridge of Spies’ into Luganda;
- Roswitha Gases, Mauricius Gariseb, Mildred Howoses, Petronella Nauises, and Dorothea Tases: translating ‘The Godfather’ and ‘Bridge of Spies’ into Khoekhoe.
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- Rania Chouia: revising and aligning ‘Bridge of Spies’ in Arabic.

⁴In ParTree, the Turkish and Hungarian texts were mixed up: the file with the Turkish ISO code contained Hungarian text, while the actual Turkish text was missing. For this release, the Turkish text was taken from ParTy, the original instantiation of the corpus.

How to cite

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